

Network Security

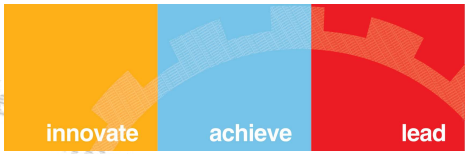
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ES ZG513/ SS ZG513, Network Security
Lecture No VI.

Mathematics for Public Key Cryptography



- Applied Cryptography
  - Number Theory Foundations
  - Modular Arithmetic, Euler & Fermat Theorems
  - Primitive Roots and Discrete Logarithm

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Why Mathematics?

- Public key systems rely on:
  - Modular arithmetic
  - Prime numbers
  - Finite cyclic groups
  - Hard mathematical problems

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Modular Arithmetic

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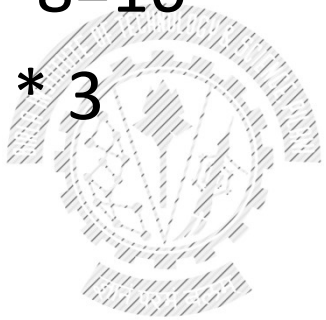
- Definition: $a \equiv b \pmod{n}$ if $n \mid (a - b)$
  - Closed under addition & multiplication
  - Efficient modular exponentiation is essential

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Fast Modular Exponentiation (Example)

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- Compute 3^13 mod 17
  - 3^2=9, 3^4=13, 3^8=16
  - 3^13 = 3^8 * 3^4 * 3
  - Answer = 12



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Euler's Totient Function

Number of integers in {1,2,…,n-1} that are coprime to n.

Important Formulas

If p is prime: $\varphi(p)=p-1$

If p prime, k≥1: $\varphi(p^k)=p^{k}-p^{k-1}$

If n = p1^e1 p2^e2: $\varphi(n)= n \Pi(1 - 1/pi)$

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Totient Example

- Compute ϕ(84)
  - 84 = 2^2 * 3 * 7
  - ϕ(84)=84(1-1/2)(1-1/3)(1-1/7)
  - Answer = 24

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Euler's Theorems

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In number theory, Euler's theorem (also known as the Fermat–Euler theorem or Euler's totient theorem) states that

If gcd(a,n)=1 then $a^{\varphi(n)} \equiv 1 \pmod{n}$ where $\varphi(n)$ is Euler's Totient Function.

Used in RSA correctness proof.

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Euler Example

- Compute 7^100 mod 40
  - $\phi(40)=16$
  - $100=16*6+4$
  - Answer = 1



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Fermat's Little Theorem

If p is prime and gcd(a,p)=1

$a^{p-1} \equiv 1 \pmod{p}$

$a^{p} \equiv a \pmod{p}$

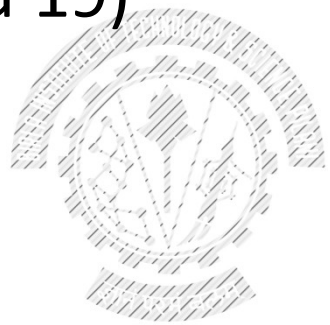


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Special case of Euler

Modular Inverse Example

- Find 5^-1 mod 19
  - 5^17 ≡ 5^-1 (mod 19)
  - Answer = 4



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RSA Mathematics

- n = pq
  - $\phi(n)=(p-1)(q-1)$
  - Choose e with $\gcd(e,\phi(n))=1$
  - Find d = e^-1 mod $\phi(n)$

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RSA Example

- p=11, q=17
  - n=187, ϕ(n)=160
  - e=7, d=23
  - Encryption/Decryption works by Euler theorem

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Primitive Roots

- Zp* is cyclic for prime p
  - Generator g has order p-1
  - Used in Diffie–Hellman

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Discrete Logarithm Problem

- Given g^x ≡ y (mod p)
  - Find x
  - Easy forward, hard reverse
  - Security basis of DH & ElGamal

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Summary

- Modular arithmetic → Efficient computation
  - Totient & Euler → RSA
  - Primitive roots → DH
  - DLP → Security foundation

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